

CLARINDA, IA, USA

WMO: 725479

Lat: 40.722N

Lon: 95.026W

Elev: 992

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	1.1	-11.2	3.1	-3.6	-6.3	4.1	4.1	26.7	27.4	24.4	26.0	4.1	0	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	93.4	77.0	91.0	76.3	89.6	75.7	80.8	89.2	78.9	87.6	77.3	85.9	9.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	155.3	86.6	76.7	144.6	84.7	74.7	134.6	82.8	45.4	89.8	43.2	87.5	41.5	84.3	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.9	20.4	18.2	DB	-15.1	97.4	7.4	3.1	-20.4	99.6	-24.8	101.4	-28.9	103.1	-34.3	105.3	(4)
			WB	-15.5	83.6	6.5	3.3	-20.1	85.9	-23.9	87.9	-27.6	89.7	-32.3	92.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	24.6	28.0	40.7	51.6	62.8	72.5	75.8	73.6	66.2	53.6	39.9	29.0
	DBStd	20.46	12.30	12.27	12.37	10.05	8.57	6.24	5.90	5.66	8.20	9.62	10.15	11.67
	HDD50	2909	787	617	337	97	8	0	0	0	2	73	329	659
	HDD65	5948	1252	1036	755	411	146	13	3	5	84	371	753	1119
	CDD50	3513	0	2	49	147	404	674	799	733	487	185	26	7
	CDD65	1077	0	0	2	11	78	237	338	273	119	17	0	2
	CDH74	11298	0	0	30	197	823	2397	3621	2632	1318	239	8	33
	CDH80	4406	0	0	3	47	256	914	1588	1041	479	62	1	15
Wind	WSAvg	7.1	7.7	8.0	9.0	9.3	8.1	6.5	5.0	4.5	5.6	7.1	7.8	7.4
Precipitation	PrecAvg	34.8	0.9	1.0	2.2	3.2	4.8	4.7	4.9	4.1	3.8	2.5	2.0	1.1
	PrecMax	52.3	3.3	3.1	6.6	7.2	9.8	10.9	14.5	15.2	11.7	6.0	5.9	2.6
	PrecMin	17.3	0.0	0.1	0.2	0.7	1.1	0.7	0.6	0.9	0.3	0.2	0.0	0.0
	PrecStd	7.2	0.7	0.8	1.5	1.8	2.2	2.3	2.8	2.8	2.4	1.5	1.4	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.6	66.0	77.4	85.9	92.6	93.6	99.1	96.7	93.1	86.4	74.9	77.1
		MCWB	46.2	52.9	59.6	64.3	70.1	75.6	76.7	77.8	73.8	68.1	59.8	72.1
	2%	DB	52.0	57.2	72.2	79.4	86.4	90.9	94.7	92.5	90.0	81.0	68.1	57.0
		MCWB	43.9	47.0	57.7	61.1	68.6	75.0	78.9	77.8	74.2	64.4	57.1	49.7
	5%	DB	45.7	50.3	66.1	74.7	81.9	88.4	91.1	89.9	85.8	75.2	62.9	50.3
		MCWB	40.3	43.4	55.3	59.7	66.1	73.8	77.8	77.2	72.0	62.2	53.4	45.0
	10%	DB	40.9	45.2	61.0	70.1	79.0	85.8	89.5	86.1	81.8	71.7	57.1	45.2
		MCWB	36.6	40.1	52.2	57.4	64.8	71.9	77.1	75.2	69.3	60.3	49.0	40.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	54.4	64.0	67.7	75.6	80.2	84.1	82.7	78.4	73.2	62.9	71.4
		MCDB	53.1	62.3	72.5	76.9	86.1	90.0	92.7	91.4	88.8	81.6	70.7	77.4
	2%	WB	44.7	48.6	60.4	64.8	71.8	77.6	81.6	80.2	76.4	69.0	58.9	52.4
		MCDB	50.3	57.3	67.9	75.2	82.2	87.9	89.6	88.8	85.8	74.6	65.0	55.2
	5%	WB	40.8	44.0	57.1	61.9	69.2	75.8	79.6	78.3	74.3	65.3	54.7	45.2
		MCDB	46.3	49.8	64.3	72.3	78.2	85.4	88.2	86.5	82.1	71.7	62.2	49.1
	10%	WB	37.1	40.3	52.5	59.1	67.0	73.9	77.8	76.6	72.0	61.1	50.0	40.6
		MCDB	40.7	45.1	61.5	67.8	75.5	83.0	86.1	84.5	79.6	68.7	57.0	45.3
Mean Daily Temperature Range	5% DB	MDBR	19.3	19.7	22.2	23.9	21.6	20.8	21.3	21.3	25.0	24.5	21.9	18.5
		MCDBR	27.1	28.5	31.9	31.8	26.4	23.4	23.9	24.6	27.8	30.4	30.7	26.0
		MCWBR	20.1	19.3	18.5	17.5	12.2	10.9	11.4	12.6	13.7	16.4	19.4	19.4
	5% WB	MCDBR	26.1	26.6	25.5	26.2	21.4	20.8	20.8	21.4	23.5	24.0	26.8	23.4
		MCWBR	19.8	19.1	16.4	16.3	11.5	11.5	12.0	12.4	13.1	16.0	18.8	19.0
Clear-Sky Solar Irradiance	taub	0.282	0.315	0.327	0.384	0.411	0.417	0.430	0.422	0.381	0.335	0.296	0.285	
	taud	2.456	2.376	2.409	2.272	2.265	2.293	2.247	2.287	2.357	2.471	2.514	2.463	
	Ebn at Noon	281	285	294	282	275	273	268	267	272	274	274	270	
	Edh at Noon	24	31	33	41	43	41	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	654	943	1223	1545	1829	2031	2039	1781	1488	1032	716	551	
	RadStd	53	61	110	96	151	167	146	100	100	118	58	56	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page